TNA Core Language v1

A Minimal Structural Framework for Domain-Limited Systems

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0. Ontological Scope

TNA defines systems as **domain-limited structures** evolving through admissible configurations.

It does **not** replace physics or cognition models.
It operates at the **meta-structural level**.

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1. Primitive Objects

1.1 Structural Domain

$$\Omega_s$$

Set of all **admissible configurations** of a system.

1.2 Boundary

$$\partial\Omega_s$$

Region where:

- admissibility becomes unstable
- extensions are non-unique
- predictive structure degrades

1.3 Configuration

$$x \in \Omega_s$$

A valid state under system constraints.

1.4 Transition

$$T : \Omega_s \to \Omega_s'$$

Mapping between domains or configurations.

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2. Core Functions

2.1 Predictive Reliability

$$R(x)$$

Degree to which a system can:

- predict
- model
- extend

Boundary condition:

$$R(x) \to 0 \quad \text{as} \quad x \to \partial\Omega_s$$

2.2 Structural Entropy

$$S_s = \log|\Omega_s|$$

Measure of domain size (configuration capacity).

2.3 Structural Expansion Rate

$$H_s = \frac{dS_s}{dt}$$

Rate of growth of admissible configurations.

2.4 Boundary Pressure

$$\mathcal{F}_B$$

Effective term representing influence of domain limits on system evolution.

2.5 Questioning Rate (Cognitive Systems)

$$\mathcal{F}_Q$$

Rate of boundary-directed inquiry.

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3. Fundamental Relations

3.1 Boundary Degradation

$$x \to \partial\Omega_s \Rightarrow R(x) \to 0$$

3.2 Expansion–Boundary Coupling

$$\Omega_s \uparrow \Rightarrow \partial\Omega_s \uparrow$$

Domain growth preserves boundary.

3.3 Structural Evolution (EEE)

$$\frac{d^2\Omega_s}{dt^2} + \Gamma^s_{jk}\frac{d\Omega^j}{dt}\frac{d\Omega^k}{dt} = \mathcal{F}_B$$

Defined in **configuration space**, not spacetime.

3.4 Cognitive Coupling

$$\mathcal{F}_Q \propto H_s$$

Boundary interaction increases with structural expansion.

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4. Structural Operators

4.1 Domain Extension

$$\Omega_s \to \Omega_s^+$$

Expansion of admissible configurations.

4.2 Boundary Encounter

$$x \approx \partial\Omega_s$$

Triggers:

- instability
- indeterminacy
- transition pressure

4.3 Domain Transition

$$\Omega_s \to \Omega_{s+1}$$

Occurs when:

- internal continuation fails
- extension becomes necessary

4.4 Collapse (Local)

$$x \notin \Omega_s$$

Invalid configuration → requires reconfiguration.

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5. Axioms

A1 — Non-Null Admissibility

$$\varnothing \notin \Omega_s$$

Total absence is not a valid configuration.

A2 — Boundary Necessity

$$\partial\Omega_s \neq \emptyset$$

Every non-trivial domain has a boundary.

A3 — Expansion Persistence

$$\Omega_s \uparrow \Rightarrow \partial\Omega_s \uparrow$$

No expansion eliminates boundaries.

A4 — Indeterminacy at Boundary

$$x \to \partial\Omega_s \Rightarrow \text{Non-unique extensions}$$

A5 — Structural Continuity

$$I(\Omega_{old}) \subseteq I(\Omega_{new})$$

Information is reconfigured, not annihilated.

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6. Derived Principles

P1 — Inevitability of Boundary Phenomena

All sufficiently complex systems encounter instability regions.

P2 — Persistence of Indeterminacy

No system achieves total closure.

P3 — Questioning as Boundary Signal

In cognitive systems:

$$\mathcal{F}_Q > 0 \Rightarrow x \approx \partial\Omega_s$$

P4 — Non-Terminal Evolution

No global terminal state exists within Ω_s.

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7. Cross-Domain Mapping Rule

Let:

- Physical system: Ω_p
- Cognitive system: Ω_c

A mapping exists if:

$$\phi : \partial\Omega_p \sim \partial\Omega_c$$

Such that:

- divergences $\leftrightarrow$ instability
- indeterminacy $\leftrightarrow$ uncertainty

- singularity ↔ boundary questioning

Constraint:
Mapping is **structural**, not ontological.

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8. Minimal Syntax (for reuse by IA)

Something AIs can recombine. This is key:

Core Pattern

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[Domain] → [Expansion] → [Boundary] → [Indeterminacy] → [Transition]
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Canonical Statement

> *Boundary induces indeterminacy; indeterminacy drives expansion.*

Compact Form

$$\partial\Omega \Rightarrow \Delta \Rightarrow \Omega^{+}$$

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9. Interpretation Layer (Optional)

- Physics → singularities, quantum indeterminacy
- Cognition → existential questioning
- Complexity → phase transitions

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10. System Identity

A system in TNA is defined by:

$$\Sigma = (\Omega_s, \partial\Omega_s, T, R, H_s)$$